

SILENT ALIASING AND CERTIFIED SAMPLING DESIGN FOR FIXED FOURIER-FEATURE MODELS

Weiping Yan

University of Minnesota, Twin Cities, Minneapolis, MN, USA

ABSTRACT

Fixed Fourier-feature coordinate models—the linear core of Fourier-feature implicit neural representations—fit a signal by least squares on an exponential dictionary Λ . We analyze, and then *design against*, the aliasing of out-of-band content. (i) **Structured indistinguishability.** A visibility/aliasability calculus shows that *arbitrary nonuniform subsets of a sampling grid inherit the grid’s aliasing equivalence classes*: an integer tone congruent mod Q to a model frequency is *exactly* indistinguishable from it—zero training residual, full-energy failure, and no estimator can avoid it. Irregular-on-a-grid sampling does *not* break exact aliasing. (ii) **Randomization breaks it, at a rate.** Off-grid jitter destroys the fold with visibility equal to the jitter characteristic function, and i.i.d. sampling gives a finite- N , dictionary-referenced concentration bound on worst-case aliasability. (iii) **Certified design.** We show the anti-aliasing objective is a D_s -optimal / null-steering criterion on sample times, give a fast surrogate whose joint objective (an ablation shows) is necessary, and—the new tool—a computable *continuum certificate* that guarantees worst-case aliasability over an entire band from the samples alone, which finite-candidate bounds and random designs do not provide. The design cuts held-out worst-case aliasability from 0.63 (random) to 0.19 ± 0.02 and certifies 0.29 band-wide vs. 0.60, in 1-D and 2-D. All closed forms are Monte-Carlo validated; trained SIREN/MLP networks are an explicitly labeled empirical extension with failures reported.

Index Terms— Sampling theory, aliasing, identifiability, optimal experimental design, Fourier features.

1. INTRODUCTION

Coordinate networks with Fourier features are standard continuous signal models [1, 2, 3]. Their analytically exact core—the object of every theorem here—is the *fixed Fourier-feature coordinate model*: a finite frequency set $\Lambda = \{\omega_1, \dots, \omega_m\} \subset \mathbb{R}$ with linear coefficients, fit by least squares. Trained nonlinear networks enter only as an *empirical extension* (Sec. 6); adaptive/learned frequencies are out of scope—we make no claim for “all INRs”.

A practitioner choosing *when* and *where* to sample faces three questions when the signal carries energy outside Λ . **Q1**: when is that energy exactly invisible in the samples, fitting perfectly yet corrupting the reconstruction? **Q2**: how does the sampling pattern change this? **Q3**: can one *design* the samples to prevent it, with a guarantee? This paper answers all three and closes the loop with a constructive, certified design.

Code, data provenance, full proofs, and a claim ledger: github.com/appleweiping/inr-aliasing-limits. Portions of the code and text were produced with AI assistance under the author’s direction (see the Compliance statement).

Contributions. (1) A visibility/aliasability calculus and its central corrective (Thm. 1): arbitrary nonuniform *grid subsets* inherit the grid’s aliasing equivalence classes, so irregular sampling on a grid does not break exact aliasing—with an estimator-independent two-point converse. (2) A finite- N , dictionary-referenced quantification of how off-grid randomization breaks the fold: a jitter characteristic-function law and an i.i.d. concentration bound (Thm. 2). (3) A *certified constructive design* (Sec. 5): we recognize the anti-aliasing objective as a D_s -optimal / null-steering criterion on sample times, give a fast joint surrogate that an ablation shows is necessary, and prove a computable *continuum aliasability certificate* that guarantees the worst case over an entire out-of-band interval from the samples only. (4) An honest empirical program: real-data both sides, a trained-network extension with its SIREN failure reported, and 2-D, all with confidence intervals and released failing runs.

Relation to prior work. Classical modulo- N aliasing [4] is the full-grid special case of Thm. 1, which covers arbitrary grid subsets; the generalized-sampling recast [5, 6] underlies our least-squares view. That randomized timing suppresses aliasing is classical—Balakrishnan’s characteristic-function attenuation and alias-free random sampling [7, 8, 9]; those are asymptotic spectral-replica statements, whereas Thm. 2 gives finite- N , dictionary-referenced control (a column-norm of the generalized aliasing operator [10]), uniform over candidates. **Our design principle is not new and we do not claim it**: protecting a model block while nulling its coupling to a specified set is exactly D_s -optimal (nuisance-parameter) experimental design [11, 12] and LCMV/MVDR null-steering [13, 14]. Our contribution is the identifiability analysis that says *which* cross-terms to null and *why grid designs cannot*, together with the *continuum certificate* (Sec. 5)—a sample-only guarantee over a whole band, which the finite-candidate bound and random designs lack. Basis mismatch [15] is the complementary near-atom regime; the Lomb–Scargle window [16] predicts periodogram leakage. The structured-dictionary view of INRs [17], INR achievability theorems [18, 19, 20], an NTK missing-frequency floor [21], anti-aliasing INR architectures [22, 23, 24], and empirical SIREN aliasing [25] are adjacent; none gives the grid-inheritance corrective, the finite- N bound, or a certified design.

2. SETUP

Signals live on $[0, 1]$; $e_\omega(x) = e^{i2\pi\omega x}$; $\mathcal{M}_\Lambda = \{\sum_k c_k e_{\omega_k}\}$ with Λ fixed. Samples $T = \{t_j\}_{j=1}^N$ give $S_T f = (f(t_j))_j$, the synthesis matrix $\Phi \in \mathbb{C}^{N \times m}$ ($\Phi_{jk} = e_{\omega_k}(t_j)$), and $\mathbf{y} = S_T f + \epsilon$. Write $f = f_{\text{in}} + f_{\text{out}}$, $f_{\text{out}} = \sum_l a_l e_{\nu_l}$, $\nu_l \notin \Lambda$. Ordinary least squares $\hat{\mathbf{c}} = \Phi^\dagger \mathbf{y}$ is used throughout. For $\nu \in \mathbb{R}$, the *sample atom* $\phi_\nu = (e^{i2\pi\nu t_j})_j$ defines the *visibility* and *aliasability*

$$v_T(\nu) = \frac{1}{\sqrt{N}} \|(\mathbf{I} - \mathbf{P}_\Lambda) \phi_\nu\| \in [0, 1], \quad a_T(\nu) = \|\Phi^\dagger \phi_\nu\|, \quad (1)$$

P_Λ the projector onto range Φ : v_T is the sample energy the model cannot absorb, a_T the coefficient bias a unit tone induces. Under white noise, $\|\hat{c} - c^*\| \leq \|\varepsilon\|/\sigma_{\min}(\Phi)$ (standard; Lemma S1). Converting coefficient error to signal error uses the continuous Gram $G_{L^2}[k, \ell] = e^{i\pi\delta} \text{sinc } \delta$ ($\delta = \omega_\ell - \omega_k$) with Riesz bounds, and $\|\hat{f} - f\|_{L^2}^2$ splits exactly into a modeled-component (aliasing-bias+noise) term, a cross term, and a truncation term $a^* G_{\text{out}} a$ that no sampling removes (Prop. S2; used in Fig. 1d and the real-data results).

3. STRUCTURED INDISTINGUISHABILITY

Theorem 1 (Exact equivalence on grid subsets). *Let $\Lambda \subset \mathbb{Z}$, T any subset of the rate- Q grid $\{0, \dots, Q-1\}/Q$ with $\sigma_{\min}(\Phi) > 0$ ($N \geq m$; Λ distinct mod Q), $\nu \in \mathbb{Z}$. (i) $v_T(\nu) = 0$ iff $\phi_\nu \in \text{range } \Phi$; if $\nu \equiv \omega_k \pmod{Q}$ then $\phi_\nu = \phi_{\omega_k}$ on every such subset, so $v_T(\nu) = 0$, the fit is the unit coefficient on atom k , and the training residual is exactly zero. (ii) With $f_1 = f_{\text{in}} + ae_\nu$, $f_2 = f_{\text{in}} + ae_{\omega_k}$ ($\nu \equiv \omega_k$, $\nu \neq \omega_k$): $S_T f_1 = S_T f_2$, so the observation laws coincide under any noise and any estimator has $\max_i \|\hat{f} - f_i\|_{L^2} \geq |a|/\sqrt{2}$. (iii) If $\nu \neq \omega_k$ for all k and $[\Phi \phi_\nu]$ has full column rank then $v_T(\nu) > 0$.*

Sketch. (i) $e^{i2\pi\nu g/Q}$ depends only on $\nu \pmod{Q}$; (ii) two-point (Le Cam) with $\|e_\nu - e_{\omega_k}\|_{L^2} = \sqrt{2}$; (iii) the rank condition (supplement). *Silent aliasing* is (i)–(ii): the samples are fit to the noise floor while the reconstruction error is $\geq$ the full out-of-band energy, at any N , with ν arbitrarily far from Λ . The corrective is (i)’s on every such subset: a common intuition holds that irregular sampling avoids aliasing; on a grid it does not, no matter how irregular the subset. Classical aliasing is $T = \text{full grid}$ [4].

4. RANDOMIZATION BREAKS IT, AT A RATE

Theorem 2 (Off-grid randomization). *Let $\Lambda \subset \mathbb{Z}$, $|\Lambda| = m$. (a) (jitter) $t_j = g_j/Q + \eta_j$, η_j i.i.d. mean-zero with characteristic function χ_η , $\nu = \omega_k + rQ$ ($r \neq 0$). For arbitrary g_j the fold coherence $\frac{1}{N} \sum_j e^{i2\pi(\nu - \omega_k)t_j}$ has mean $\chi_\eta(2\pi rQ)$ and concentrates at rate $2\sqrt{\log(4/\delta)/N}$; if the g_j are drawn i.i.d. uniformly and Λ is distinct mod Q then $v_T(\nu)^2 = 1 - |\chi_\eta(2\pi rQ)|^2 + O_P(m\sqrt{\log(m/\delta)/N})$, giving (Gaussian σ_t) $v_T(\nu) \approx 2\pi|r|Q\sigma_t$. (b) (i.i.d.) If the limiting normalized Gram is $\succeq \lambda_{\min} \mathbf{I}$ and limiting model-candidate coherences vanish on a finite Ω , $|\Omega| = K$ ($\lambda_{\min} = 1$ for the uniform density and integer-disjoint candidates), then with $\varepsilon = 2\sqrt{\log(4(mK + m^2)/\delta)/N}$, $m\varepsilon < \lambda_{\min}$, w.p. $\geq 1 - \delta$, $\max_{\nu \in \Omega} a_T(\nu) \leq \sqrt{m\varepsilon}/(\lambda_{\min} - m\varepsilon)$.*

Sketch. (a) the on-grid fold phase is 1; jitter adds i.i.d. unit-modulus factors (empirical characteristic function; Hoeffding). (b) Hoeffding+union over $mK + m^2$ coherences, then a Neumann/perturbation bound on $a_T(\nu) = \|(\Phi^* \Phi/N)^{-1}(\Phi^* \phi_\nu/N)\|$; the constants are loose (a union bound) but the $N^{-1/2}$ rate is exact (Fig. 1a,b). Two honest boundaries: on-grid designs never escape Thm. 1; and (b) controls only a finite Ω —the certificate of Sec. 5 removes that limitation.

5. CERTIFIED ANTI-ALIASING SAMPLING DESIGN

Given Λ , a candidate out-of-band set Ω the practitioner wants visible, and a budget N , we choose sample times to make Φ well conditioned and $\Phi^* \Psi_\Omega$ small (Ψ_Ω the Ω atoms): jointly, high

$\min_\nu v_T(\nu)$ and low $\max_\nu a_T(\nu)$ at small $\kappa(\Phi)$. This asymmetric block target—protect $\Phi^* \Phi \rightarrow \mathbf{I}$, null $\Phi^* \Psi_\Omega \rightarrow 0$, ignore the Ω self-block—is precisely D_s -optimal (nuisance-parameter) design [11, 12] and LCMV null-steering [13, 14]; we credit those and add the identifiability reading (which cross-terms, and why grid designs fail: Thm. 1) and a certificate.

Surrogate and ablation. The exact objective is non-smooth and (via $\Phi^\dagger$) non-submodular—we claim no greedy guarantee. We minimize $J(T) = \frac{1}{N^2} \sum_{k,\nu} |(\Phi^* \Psi_\Omega)_{k\nu}|^2 + \beta \frac{1}{N^2} \sum_{k \neq l} |(\Phi^* \Phi)_{kl}|^2$, a function of the difference-frequency sums, by coordinate descent over continuous times (fast rank-one updates; deterministic given init; a grid-restricted greedy variant exists but inherits Thm. 1 and cannot break exact folds). Both terms are needed: coherence-only design nulls aliasing but wrecks conditioning ($\kappa \approx 2.1$), condition-only design keeps $\kappa \approx 1.1$ but leaves $\max a_T \approx 0.53$; only the joint objective gets both (Fig. 2a).

Continuum certificate. $a_T(\nu)^2 = \phi_\nu^* M \phi_\nu$, $M = \Phi(\Phi^* \Phi)^{-2} \Phi^*$, is a real trigonometric polynomial in ν with frequencies $t_l - t_j$; by a Bernstein/Lipschitz bound (its derivative is $\leq 2\pi \sum_{j,l} |M_{jl}| |t_l - t_j|$) a fine grid plus a slack term certifies $\max_{\nu \in [a,b]} a_T(\nu)$ over an entire continuous band from the samples alone (Prop. 1; sound and tight in the tests). This is exactly what Thm. 2(b) cannot give (finite Ω) and random designs have no analogue.

Proposition 1 (Certificate). *For any T , $\max_{\nu \in [a,b]} a_T(\nu) \leq (\max_{\text{grid}} \phi^* M \phi + Lh/2)^{1/2}$ with grid spacing h and $L = 2\pi \sum_{j,l} |M_{jl}| |t_l - t_j|$; an analogous bound lower-bounds $\min_{\nu \in [a,b]} v_T(\nu)$.*

Result. Designed on a focused, thickened concern set and evaluated on held-out perturbed frequencies it never saw, the design cuts worst-case aliasability to 0.19 ± 0.02 vs. 0.63 (random jitter) and 0.58 (E-optimal), and stays at 0.20 under candidate-set misspecification (Fig. 2b); the advantage persists across the budget sweep (Fig. 2c) and translates to lower signal-domain aliasing bias at equal noise gain (Fig. 2d). Over a continuous band it certifies $\max a_T \leq 0.29$ vs. 0.60 for random. It generalizes to 2-D frequency vectors ($\max a_T$ 0.17 vs. 0.68). Honest scope: the concern set must be focused ($K = O(N)$); for a broad band densely covered with $K \gg N$ no design beats random much, and a badly misspecified set loses the edge.

6. EXPERIMENTS

All closed forms are Monte-Carlo validated (39 tests; CI on every push). JSONs carry seeds, configs, and environment stamps; every figure is regenerated from them.

Theory (Fig. 1). (a) the jitter law matches $\sqrt{1 - \chi_\eta^2}$ across two decades, exact fold ($\nu \approx 10^{-14}$) at $\sigma_t=0$; (b) i.i.d. worst-case aliasability decays $N^{-1/2}$ beneath the bound; (c) the grid-subset coherent tone stays invisible at every N while i.i.d. sampling makes even the per- N adversarial candidate visible; (d) jitter removes the aliasing-bias term, never truncation.

Design (Fig. 2). The four panels above; the joint objective, held-out generalization, budget sweep, and signal-domain payoff.

Real signals, sample-only. On CO₂, sunspots, and nine speech segments (anti-aliased decimation; full records), sample-only Lomb–Scargle bandwidth selection attains RMSE 0.167 vs. an evaluation-only oracle 0.161 on CO₂; on broadband speech no selector beats RMSE ≈ 1.00 —the truncation term dominates at that budget (the real-data face of the converse). Details, the

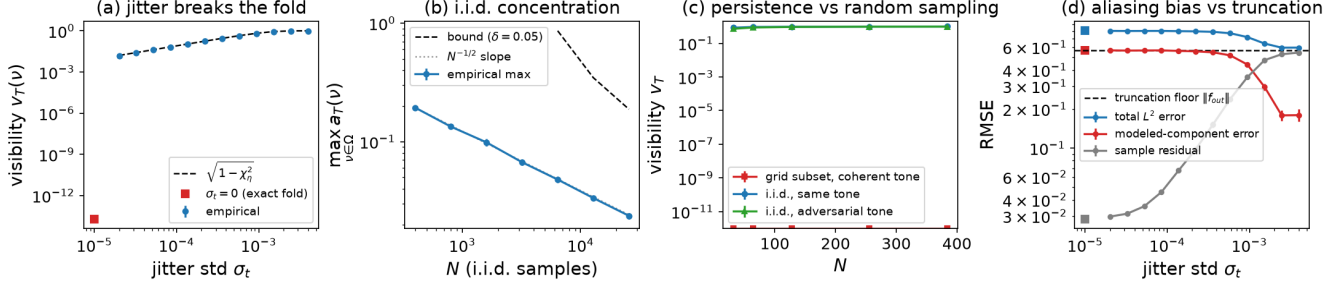


Fig. 1. Theory validation (mean $\pm$ 95% CI, 20 draws). (a) jitter law $\sqrt{1 - \chi^2}$; (b) i.i.d. worst-case aliasability vs. the bound; (c) grid persistence vs. i.i.d. (fixed *and* adversarial tones); (d) jitter removes aliasing bias, not truncation.

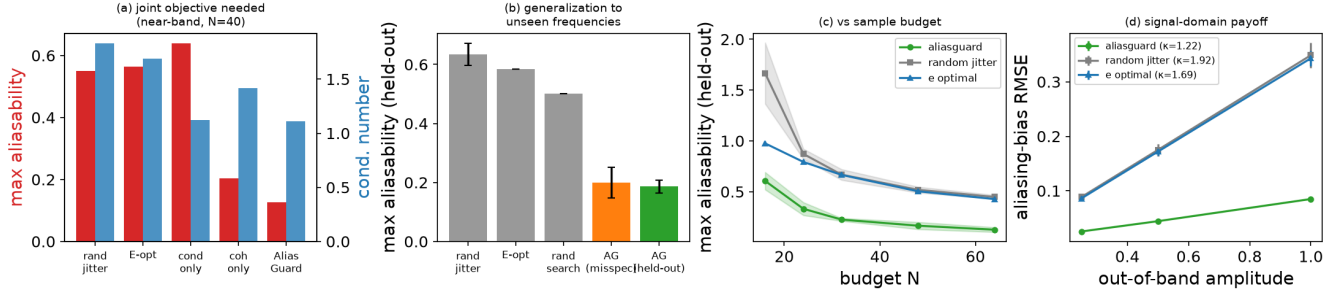


Fig. 2. Certified anti-aliasing design. (a) ablation—only the joint objective gets both low aliasability *and* low condition number (near-band, $N=40$); (b) generalization to held-out, unseen frequencies (mean $\pm$ 95% CI), incl. under candidate-set misspecification; (c) advantage across the sample budget; (d) signal-domain payoff: lower aliasing-bias RMSE at equal noise gain κ .

detection dichotomy (a coherent fold is indistinguishable from its in-band twin by any sample-only test; visible tones follow an exact noncentral- χ^2 power curve matched by calibrated detectors), and a trained-network extension (FF-MLP fold match 77%/corr 0.96 over ≥ 20 seeds; *SIREN fails*, corr 0.49—trained SIRENs leave their init NTK, so the theory makes no claim there) with 2-D results are in the supplement, with all failing runs released.

7. LIMITATIONS AND CONCLUSION

The theory covers fixed-feature linear-coefficient models; nonlinear networks are an empirical extension that *fails for SIRENs*, and adaptive frequencies are open. The concentration constants are loose (union bound); the design needs a focused concern set. Within that scope: irregular-on-a-grid sampling does not break exact aliasing (Thm. 1); off-grid randomization does, at an explicit finite- N rate (Thm. 2); and a D_s -optimal / null-steering design with a continuum certificate turns the analysis into a deployable, guaranteed anti-aliasing sample pattern. Practical guidance: never sample purely on a grid when out-of-band content is possible; target a focused concern set; and certify the band, not a finite list.

Compliance with Ethical Standards. This is a theoretical and computational study; it used only publicly available data (Mauna Loa CO₂, NOAA; sunspot numbers, SILSO/Royal Observatory of Belgium; Open Speech Repository Harvard sentences), under those sources’ terms, and involved no human or animal subjects. The authors declare no conflicts of interest and no external funding. Code and text were produced with AI assistance (Anthropic Claude) under the author’s direction; all results were generated by the released, reproducible code.

8. REFERENCES

- [1] V. Sitzmann, J. Martel, A. Bergman, D. Lindell, and G. Wetzstein, “Implicit neural representations with periodic activation functions,” in *NeurIPS*, 2020.
- [2] M. Tancik, P. Srinivasan, B. Mildenhall, et al., “Fourier features let networks learn high frequency functions in low dimensional domains,” in *NeurIPS*, 2020.
- [3] B. Mildenhall, P. Srinivasan, M. Tancik, et al., “NeRF: Representing scenes as neural radiance fields for view synthesis,” in *ECCV*, 2020.
- [4] C. E. Shannon, “Communication in the presence of noise,” *Proc. IRE*, vol. 37, no. 1, pp. 10–21, 1949.
- [5] B. Adcock and A. C. Hansen, “Generalized sampling and infinite-dimensional compressed sensing,” *Found. Comput. Math.*, vol. 16, pp. 1263–1323, 2016.
- [6] M. Unser, “Sampling—50 years after Shannon,” *Proc. IEEE*, vol. 88, no. 4, pp. 569–587, 2000.
- [7] A. V. Balakrishnan, “On the problem of time jitter in sampling,” *IRE Trans. Inform. Theory*, vol. 8, no. 3, pp. 226–236, 1962.
- [8] H. S. Shapiro and R. A. Silverman, “Alias-free sampling of random noise,” *J. Soc. Indust. Appl. Math.*, vol. 8, no. 2, pp. 225–248, 1960.
- [9] F. J. Beutler, “Alias-free randomly timed sampling of stochastic processes,” *IEEE Trans. Inform. Theory*, vol. 16, no. 2, pp. 147–152, 1970.

- [10] M. K. Transtrum, G. Hart, T. J. Jarvis, and J. P. Whitehead, “eGAD! Double descent is explained by generalized aliasing decomposition,” *arXiv:2408.08294*, 2024.
- [11] S. D. Silvey, *Optimal Design*, Chapman & Hall, 1980.
- [12] A. C. Atkinson, A. N. Donev, and R. D. Tobias, *Optimum Experimental Designs, with SAS*, Oxford Univ. Press, 2007.
- [13] O. L. Frost, “An algorithm for linearly constrained adaptive array processing,” *Proc. IEEE*, vol. 60, no. 8, pp. 926–935, 1972.
- [14] H. L. Van Trees, *Optimum Array Processing*, Wiley, 2002.
- [15] Y. Chi, L. L. Scharf, A. Pezeshki, and A. R. Calderbank, “Sensitivity to basis mismatch in compressed sensing,” *IEEE Trans. Signal Process.*, vol. 59, no. 5, pp. 2182–2195, 2011.
- [16] J. T. VanderPlas, “Understanding the Lomb–Scargle periodogram,” *Astrophys. J. Suppl. Ser.*, vol. 236, no. 1, 2018.
- [17] G. Yüce, G. Ortiz-Jiménez, B. Besbinar, and P. Frossard, “A structured dictionary perspective on implicit neural representations,” in *CVPR*, 2022.
- [18] M. Najaf and G. Ongie, “Towards a sampling theory for implicit neural representations,” in *Asilomar Conf. Signals, Syst., Comput.*, 2024, arXiv:2405.18410.
- [19] M. Najaf and G. Ongie, “Sampling theory for super-resolution with implicit neural representations,” *SIAM J. Imaging Sci.*, 2025, arXiv:2506.09949.
- [20] H. Saratchandran, S. Ramasinghe, V. Shevchenko, A. Long, and S. Lucey, “A sampling theory perspective on activations for implicit neural representations,” in *ICML*, 2024, arXiv:2402.05427.
- [21] K. Ma, M. Zhu, Y. Zhan, H. Yin, S. Wang, and Y. Zheng, “Robustifying Fourier features embeddings for implicit neural representations,” *arXiv:2502.05482*, 2025.
- [22] D. B. Lindell, D. Van Veen, J. J. Park, and G. Wetzstein, “BACON: Band-limited coordinate networks for multiscale scene representation,” in *CVPR*, 2022.
- [23] J. Barron, B. Mildenhall, M. Tancik, et al., “Mip-NeRF: A multiscale representation for anti-aliasing neural radiance fields,” in *ICCV*, 2021.
- [24] A. Shabanov, S. Govindarajan, C. Reading, et al., “BANF: Band-limited neural fields for levels of detail reconstruction,” in *CVPR*, 2024.
- [25] H. Chandravamsi et al., “Spectral bottleneck in sinusoidal representation networks: Noise is all you need,” *arXiv:2509.09719*, 2025.